Graph Neural Networks For Multivariate Time Series Activity Recognition

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Contents

Introduction

A controlled benchmark of temporal graph neural networks
and graph-construction strategies

The channels are not independent

- Several signals, one clock, physically coupled
  - on the body the coupling is anatomy
- LSTM · TCN · CNN–BiGRU model time well
  - but consume the channels as an unordered vector
- A graph makes the relationship explicit
  - nodes are channels, edges are relationships, message passing every step

The adjacency is not given

- **A temporal graph model cannot run until an adjacency is supplied**

– traffic and molecular graphs are observed, a sensor set has none

- **It must therefore be constructed, by one of three routes**

– domain knowledge · statistical estimation · end-to-end learning

- **In the literature the choice is made once and reported in one sentence**

– so graph and architecture vary together, and neither can be attributed

Problem Statement

Graph construction is an unmeasured free variable of every published temporal graph result.

- **Methodological**

– varying graph and architecture together permits no attribution

- **Practical**

– the choice must be made, and there is no evidence base for it

- **Scientific**

– predefined -vs- no-graph confounds the domain knowledge with the aggregation

Research questions

- **RQ1**

– Does a graph beat sequence models — and is the answer conditional on which graph?

- **RQ2**

– Which properties of a graph govern performance, and what is each worth?

- **RQ3**

– Does the domain knowledge contribute anything beyond the aggregation?

A–B answer RQ1 · C–D answer RQ2 · E–H answer RQ3

Related work

Study	Contribution	Where its graph comes from
Seo et al. 2018 · GConvLSTM	Graph convolutions inside LSTM / GRU cells	given — fixed, never varied
Li et al. 2018 · DCRNN	Diffusion convolution in a GRU	given — the road network
Bai et al. 2021 · A3TGCN	Temporal attention over graph-conv GRUs	given — the road network
Yu et al. 2018 · STGCN	Interleaved graph and temporal convolution	given — the road network
Yan et al. 2018 · ST-GCN	Skeleton-based action recognition	observed — the skeleton
Shi et al. 2019 · 2s-AGCN	Makes the skeleton adjacency partly learnable	observed, then adapted
Wu et al. 2019 · Graph WaveNet	Self-adaptive adjacency trained end-to-end	learned — inside one design
Wu et al. 2020 · MTGNN	Graph-learning module with sparsity control	learned — inside one design
Ordóñez & Roggen 2016 · DeepConvLSTM	The wearable-HAR reference architecture	no graph at all

In every one the graph is given, observed, or learned inside one proposed design — never varied as the object of study.

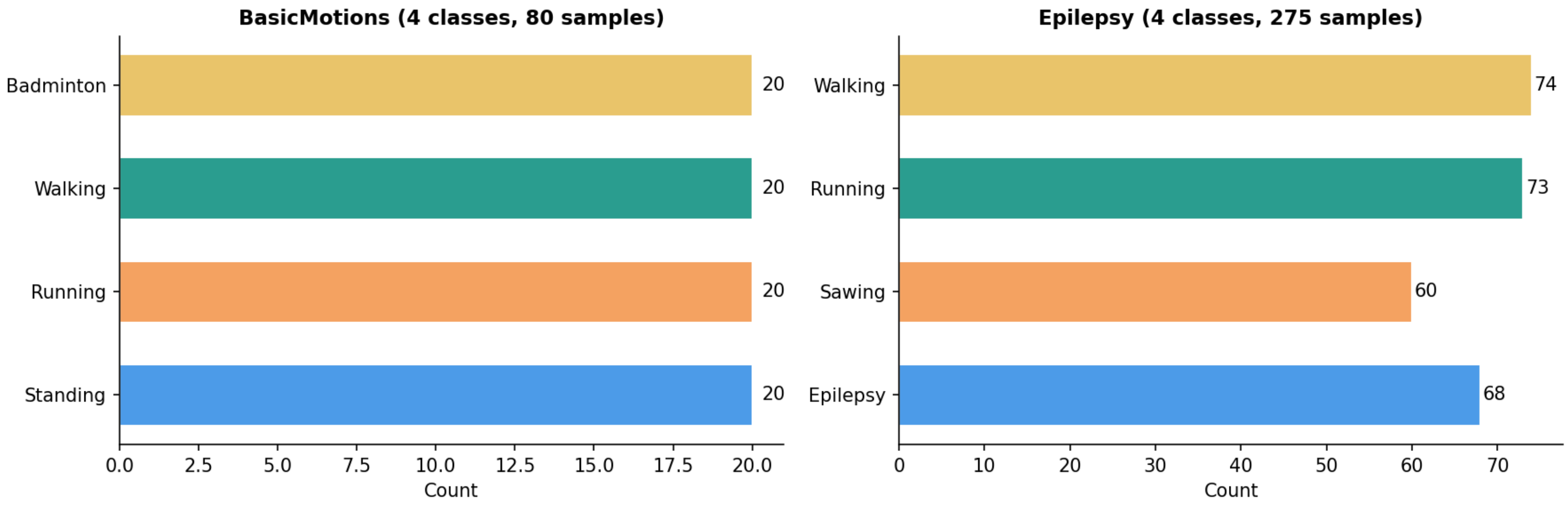
Data

Phase 1 — the two UEA datasets

Attribute	BasicMotions	Epilepsy
Source	UEA Archive	UEA Archive
Classes	4 (Standing / Walking / Running / Badminton)	4 (Epilepsy / Walking / Running / Sawing)
# Samples	80 (20 per class)	275 (≈ 69 per class)
Channels	6 (AccX/Y/Z + GyroX/Y/Z)	3 (AccX/Y/Z)
Timesteps	100 (10 Hz → 10 s)	206 (16 Hz → ~13 s)
Split	40 train / 40 test	137 train / 138 test
Normalisation	zero-mean, unit-variance per channel	zero-mean, unit-variance per channel

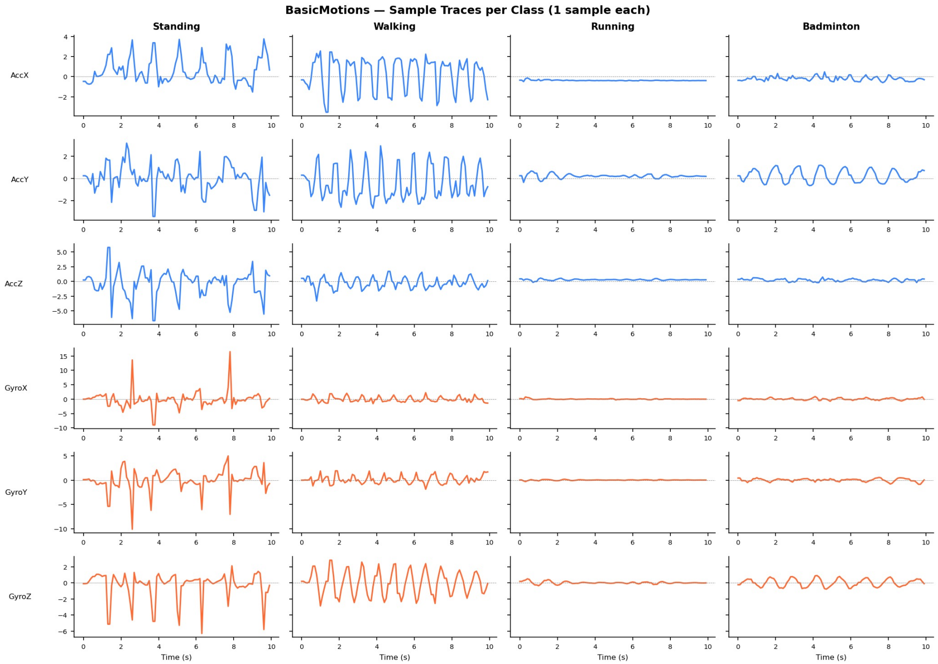
Both are single wrist-worn units on the archive's own fixed split.

Class distributions



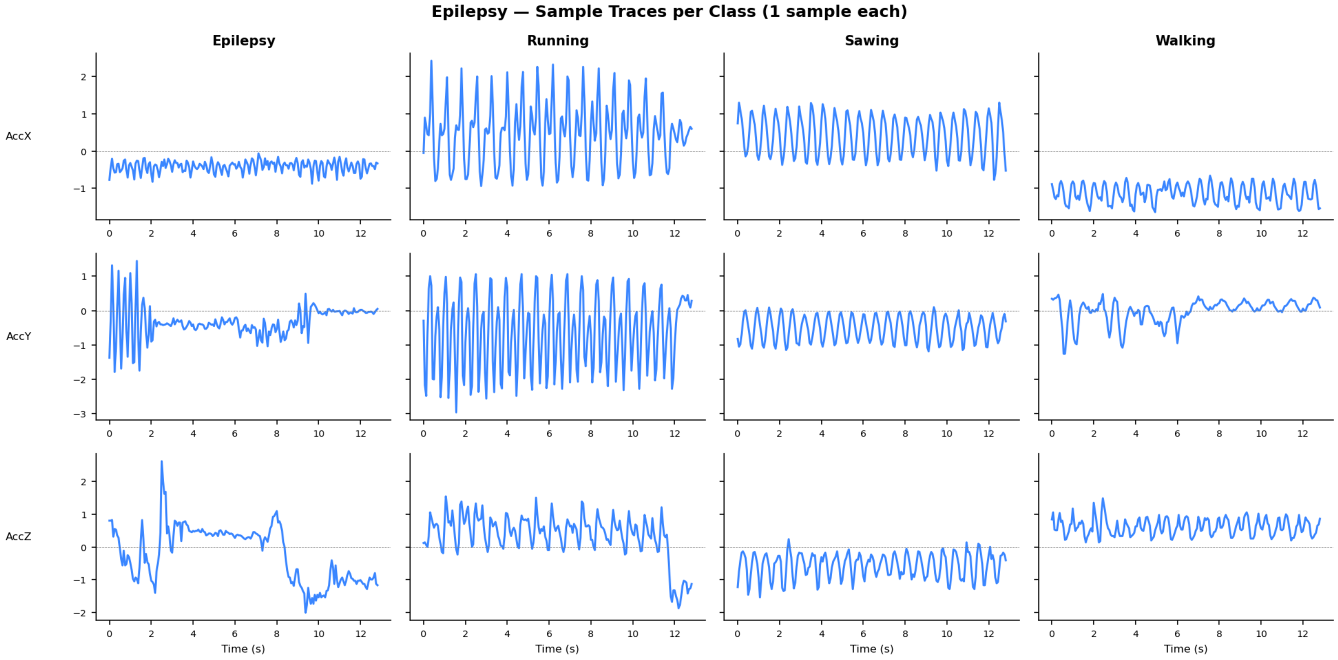
BasicMotions and DSA are exactly balanced. Epilepsy is mildly imbalanced (60 sawing → 74 walking windows).

BasicMotions — what the signals look like



Six channels, one window per class, before normalisation.

Epilepsy — what the signals look like

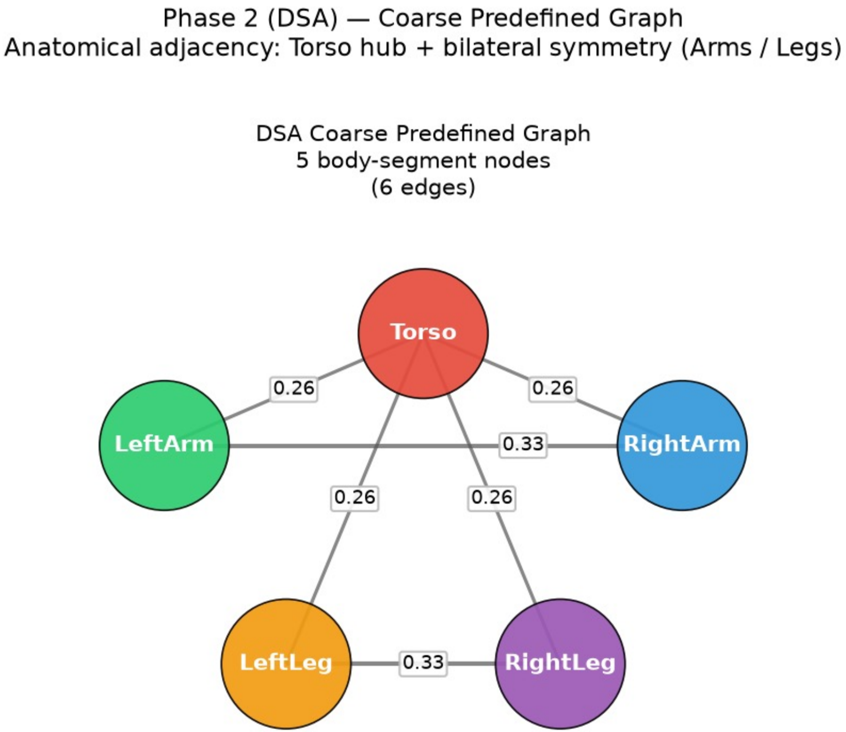


Three accelerometer axes, one window per class, before normalisation.

DSA — what the signals look like

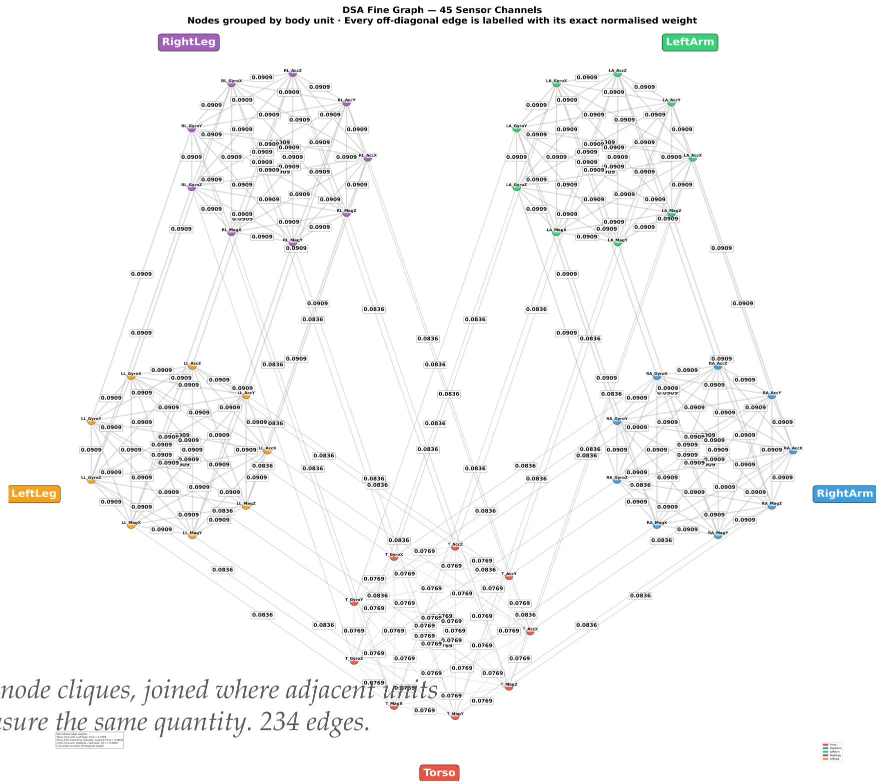
Torso unit, 9 channels, 6 activities, 3 subjects. Raw units.

The coarse graph — 5 body-segment nodes



Torso as hub, two bilateral edges. 6 of 10 possible edges.

The fine graph — 45 channel nodes



Five nine-node cliques, joined where adjacent units measure the same quantity. 234 edges.

Three datasets, chosen against requirements

Dataset	Phase	Ch.	Inst.	Cls	Window	Protocol
BasicMotions	1	6	80	4	100 @ 10 Hz = 10 s	fixed archive split
Epilepsy	1	3	275	4	206 @ 16 Hz = 13 s	fixed archive split
DSA	2	45	9,120	19	125 @ 25 Hz = 5 s	8-fold leave-one-subject-out

- 3 → 45 channels, so transfer can be tested
- DSA alone has real anatomy and multiple subjects

– its 45 channels organise as 45 nodes OR 5 body-segment nodes

Methodology

The controlled variable

- **Predefined**

– anatomy — coarse 5-node, fine 45-node

- **Data-driven**

– correlation · DTW · mutual kNN

- **Adaptive**

– learned through the process, with or without SVD seeding

- **Random-topology control**

– same nodes, same edge count, edges placed at random

Coarse against fine is a reshape, not an average

$$(T, 45) \quad \rightarrow \quad (T, 5 \text{ nodes} \times 9 \text{ features})$$

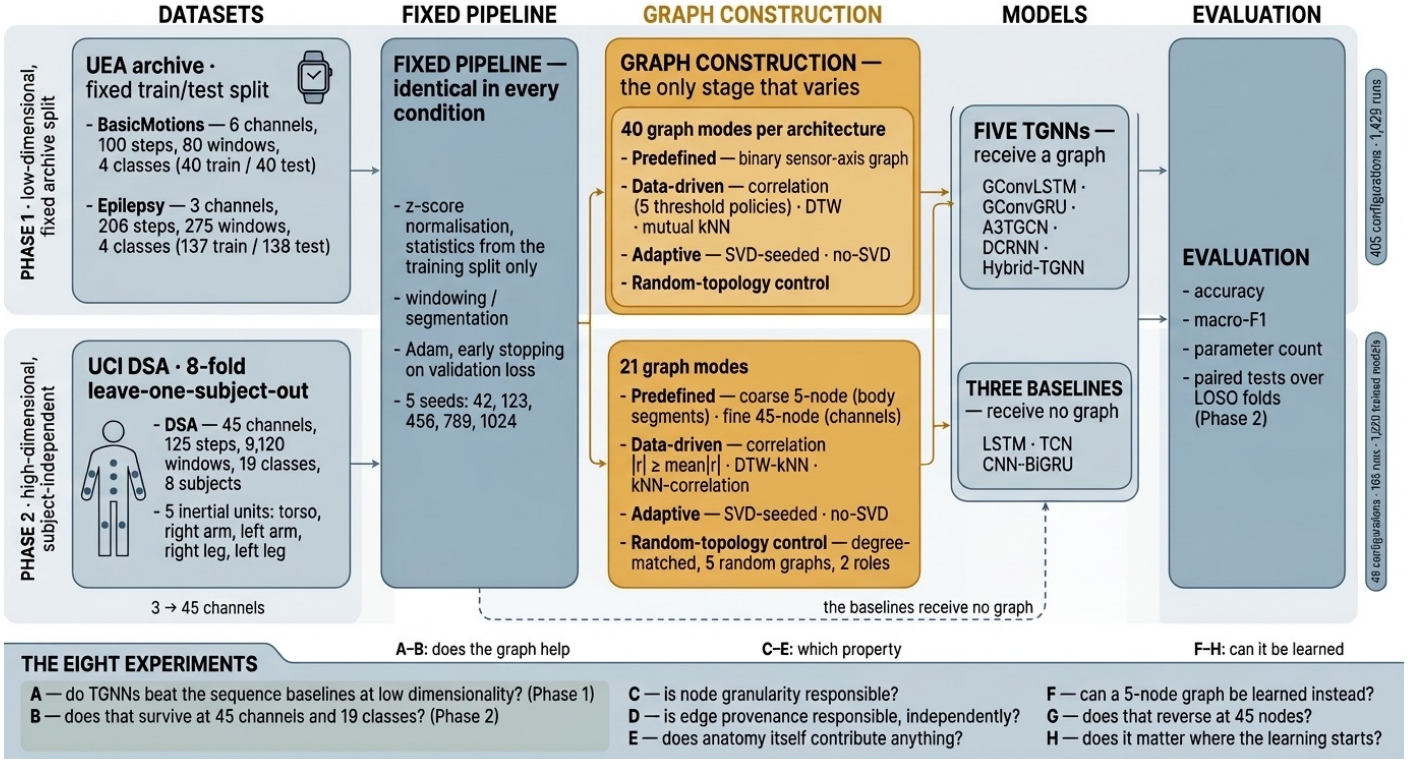
- **No channel is discarded, the tensor is reorganized, not reduced.**

– Fine is the same data as 45 nodes × 1 feature.

- **The pair therefore isolates node granularity from every other factor.**

Experimental design

The framework



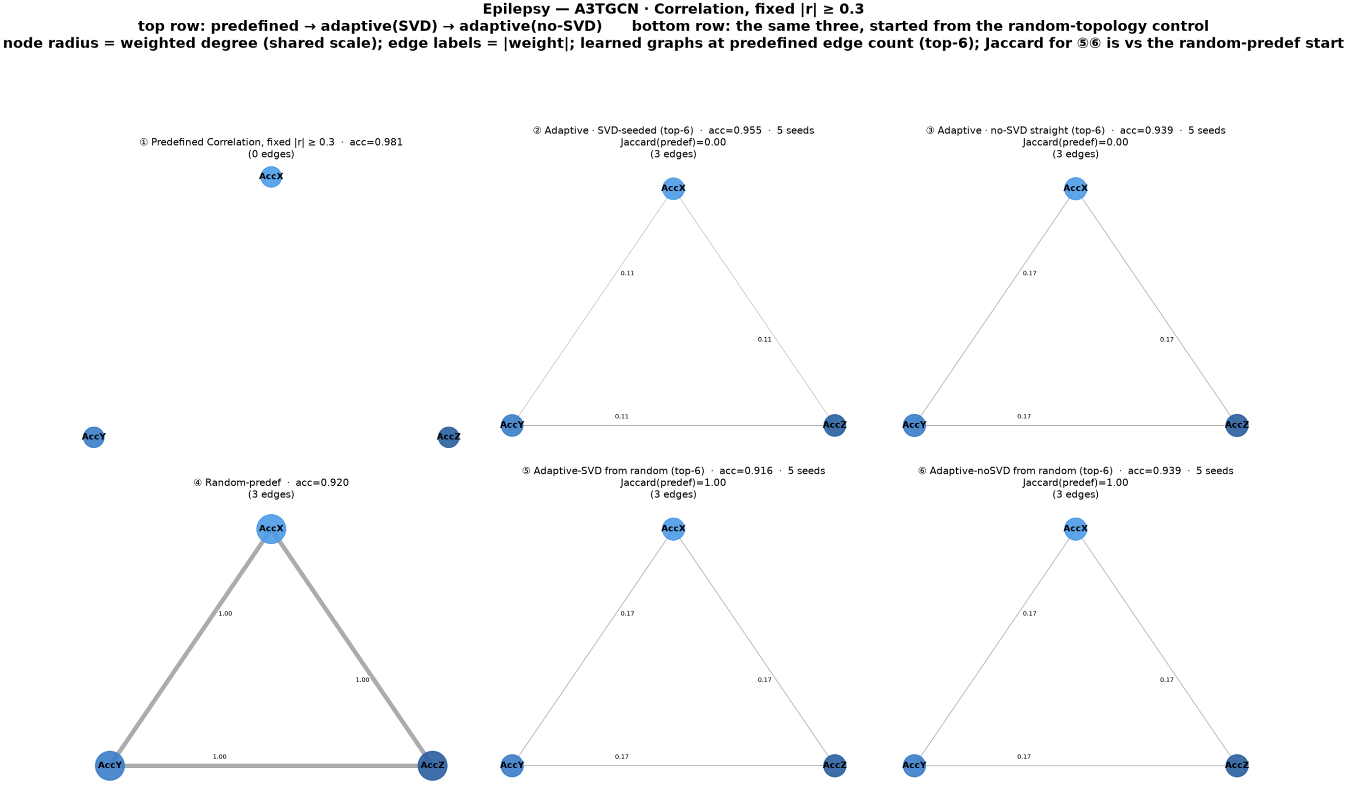
Everything is held fixed except graph construction.

What each model was given

- **Graphs reconstructed from trained checkpoints**
  – predefined → adaptive(SVD) → adaptive(no-SVD), vs random control
- **Accuracy tracks the graph SOURCE**
- **At 3–6 nodes the learner barely moves**
  – Jaccard ≈ 1 — no room for learning to help

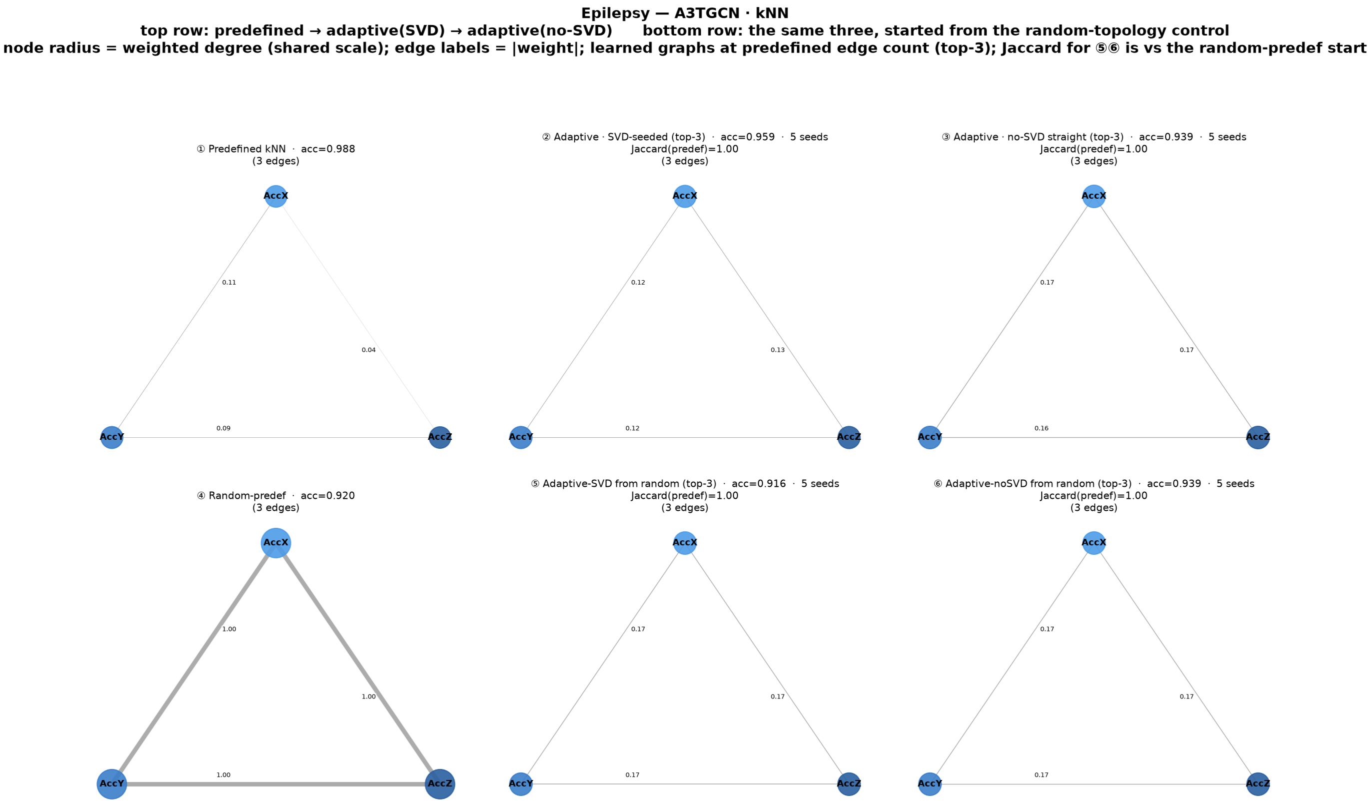
Phase 1 results

Epilepsy — the fixed-threshold correlation graph



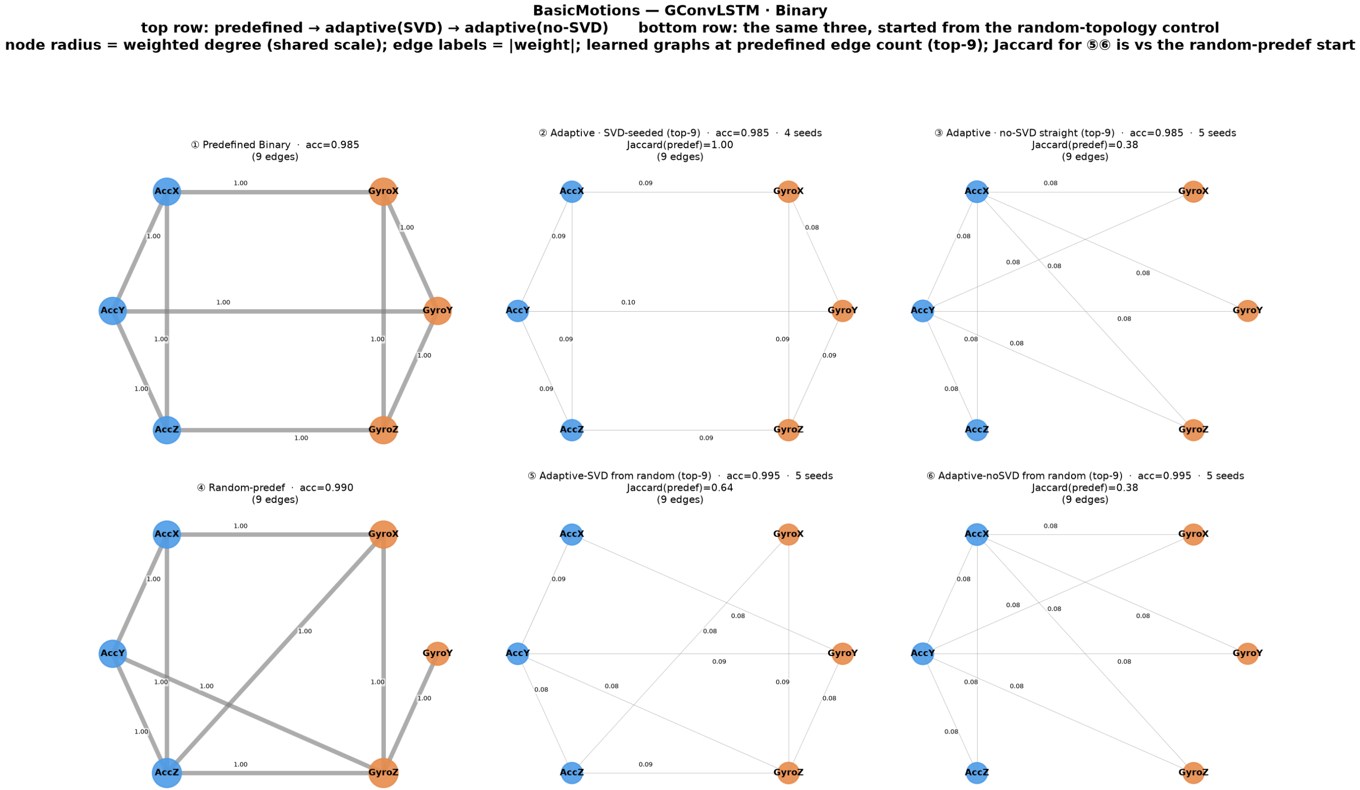
Panel ① is EMPTY. Three nodes, zero edges — every |r| falls below 0.3.

Epilepsy — the kNN data-driven graph



98.8 % as a PREDEFINED graph. Learning on top lowers it.

BasicMotions — six sensor axes



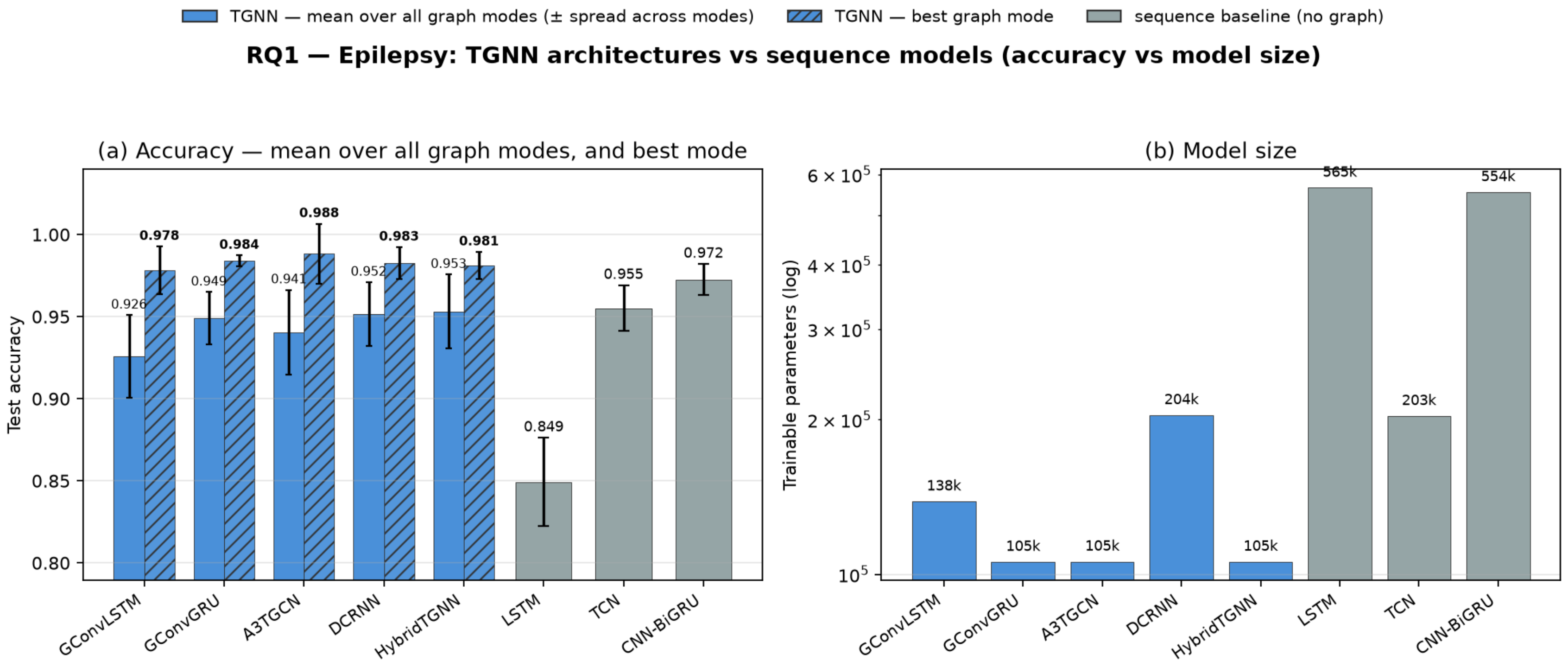
GConvLSTM. 9 of 15 possible edges — not fully connected.

Experiment A — Phase 1

Epilepsy	Accuracy	Adjacency
TGNNs, predefined axis graph	91.8	lose by 5.4
TGNNs, data-driven graph	97.4	parity
CNN–BiGRU (strongest baseline)	97.2	554 k params
Lightest TGNNs	—	~105 k (5× smaller)

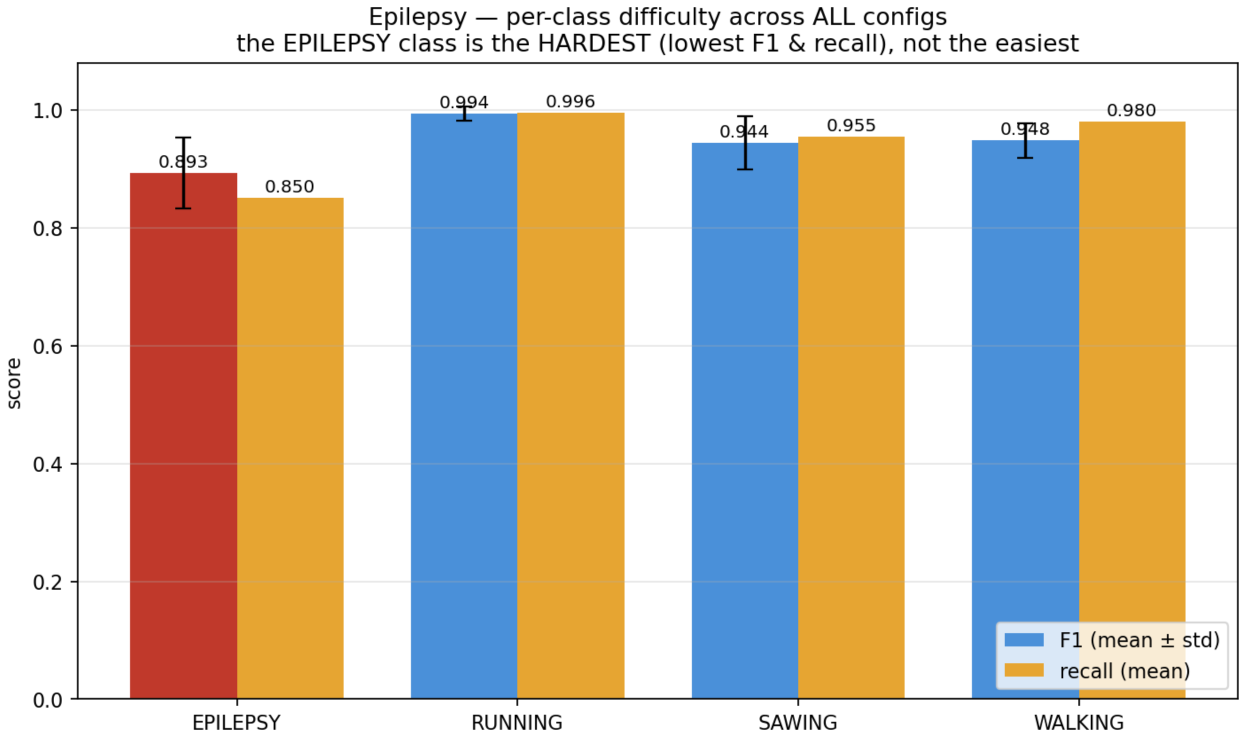
Same five architectures. Only the adjacency changed.

Phase 1 accuracy against cost



Solid = mean over all 40 graph modes; hatched = best mode.

Accuracy is not the whole result — per-class difficulty



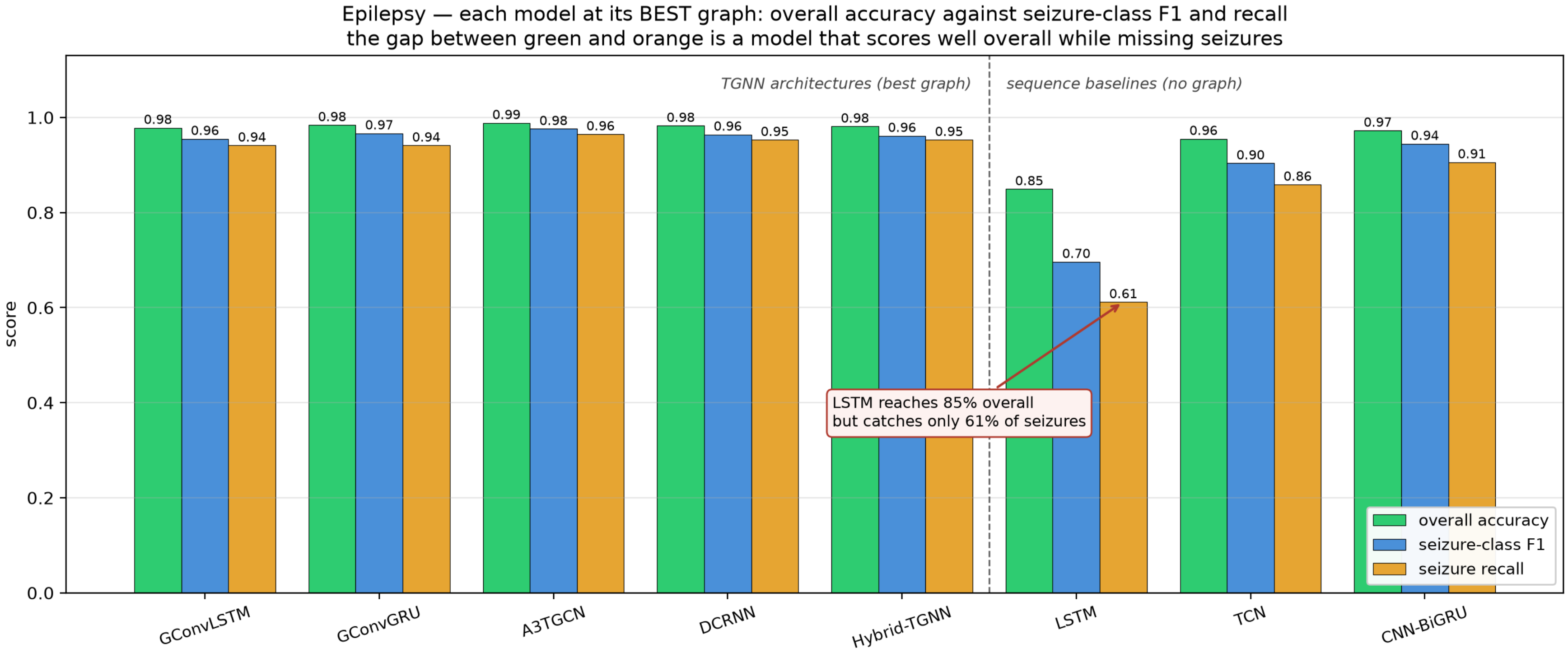
Averaged over every configuration. The seizure class is the hardest, not the easiest.

What accuracy conceals on the seizure class

Epilepsy, seizure class	Recall	F1
LSTM	0.612	0.696
TCN	0.859	0.904
CNN--BiGRU (strongest baseline)	0.906	0.945
GConvLSTM / GConvGRU	0.941	0.955 / 0.967
DCRNN / Hybrid-TGNN	0.953	0.964 / 0.961
A3TGCN	0.965	0.976

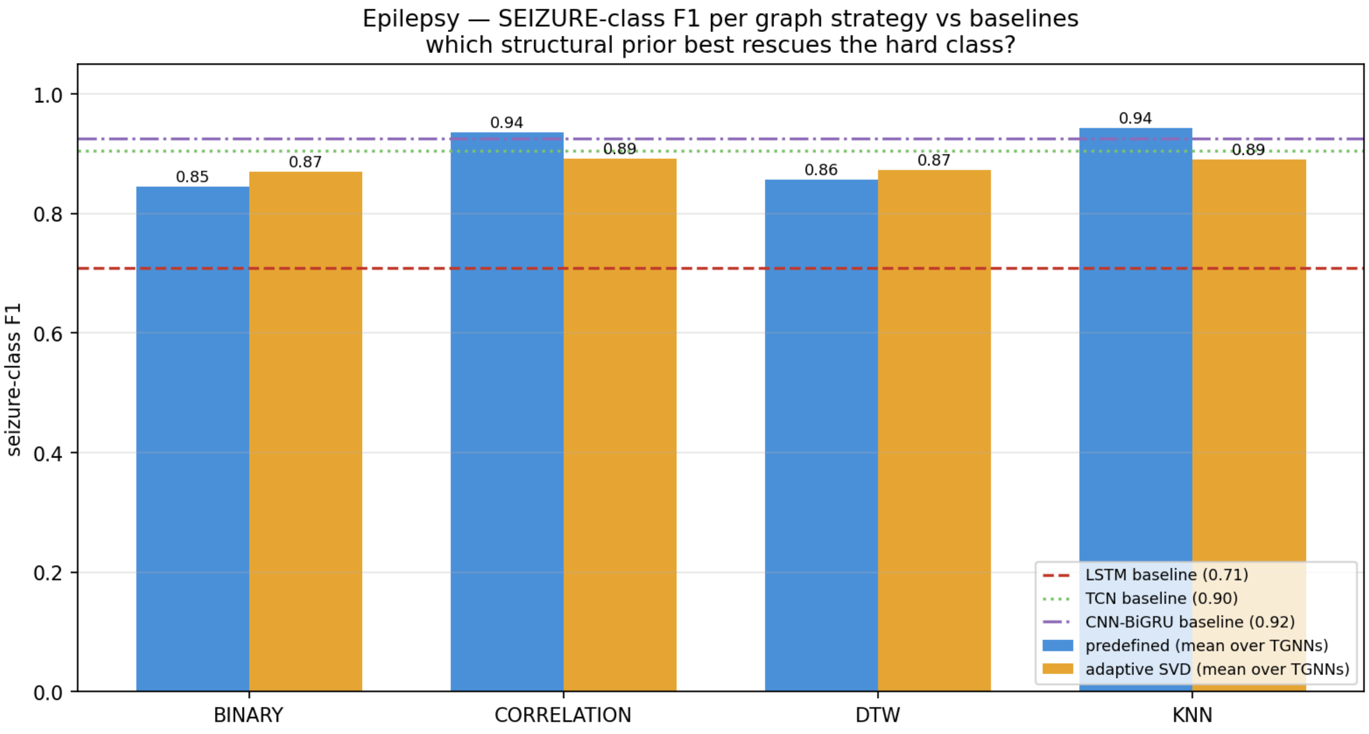
The LSTM misses 39 % of seizures. A3TGCN misses 3.5 %.

Overall accuracy against seizure performance



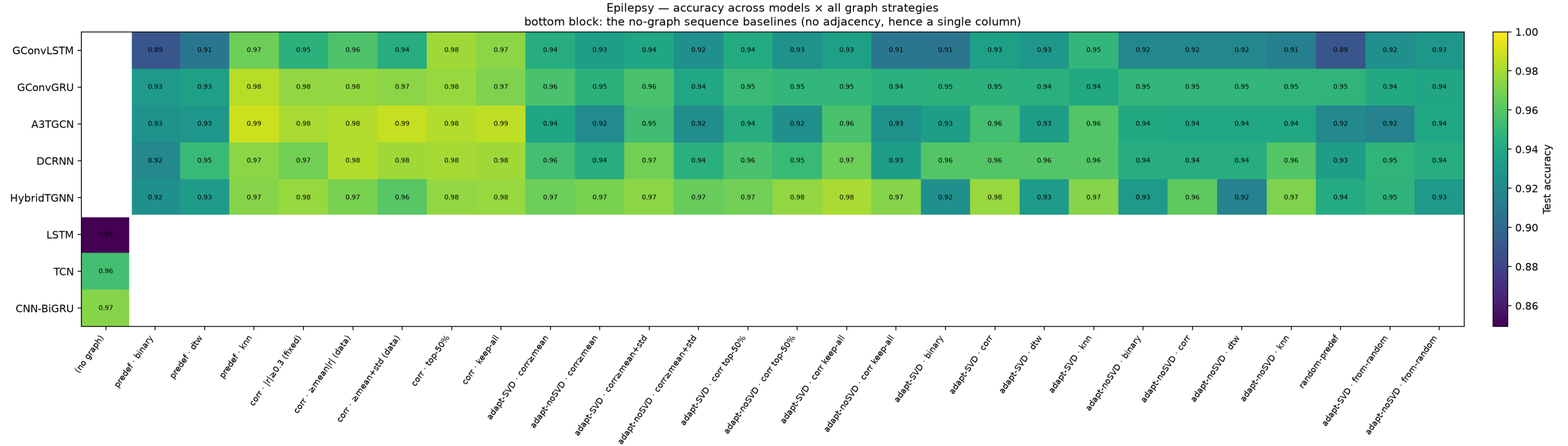
The LSTM misses 39 % of seizures. A3TGCN misses 3.5 %.

Which construction rescues the seizure class



Seizure-class F1 by graph strategy, against the three baselines.

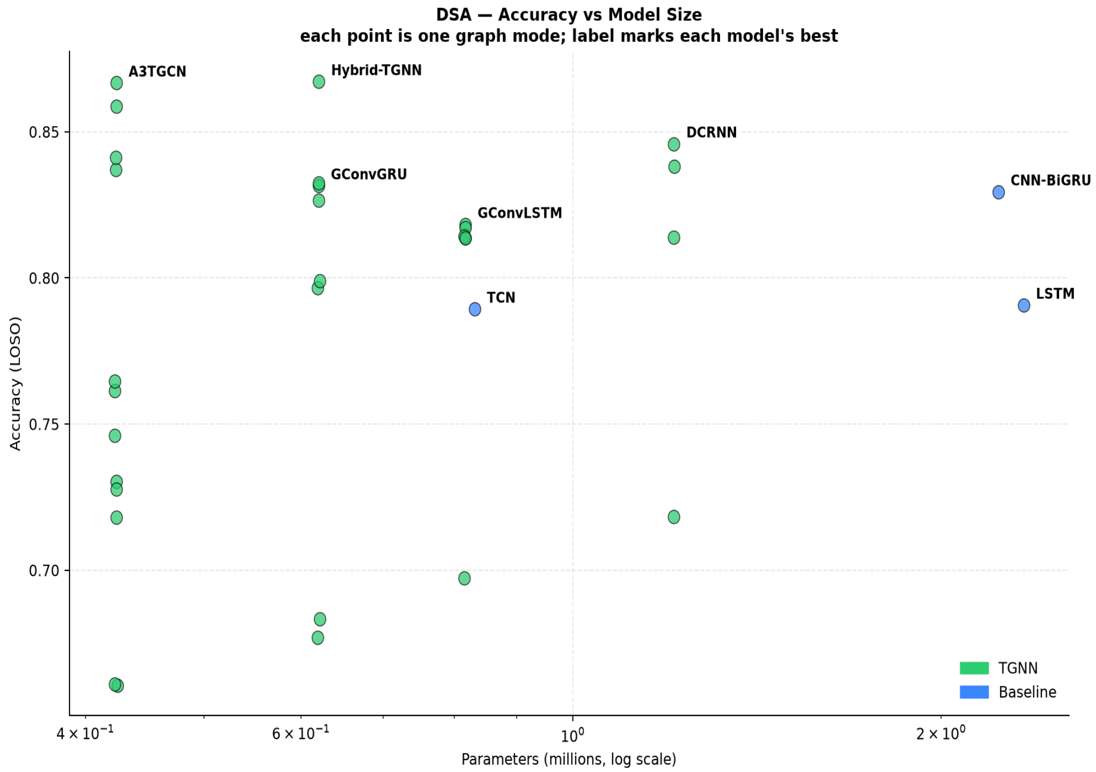
Every architecture × every strategy



Epilepsy. Rows are architectures, columns are the 40 graph type modes.

Phase 2 results

Accuracy against model size



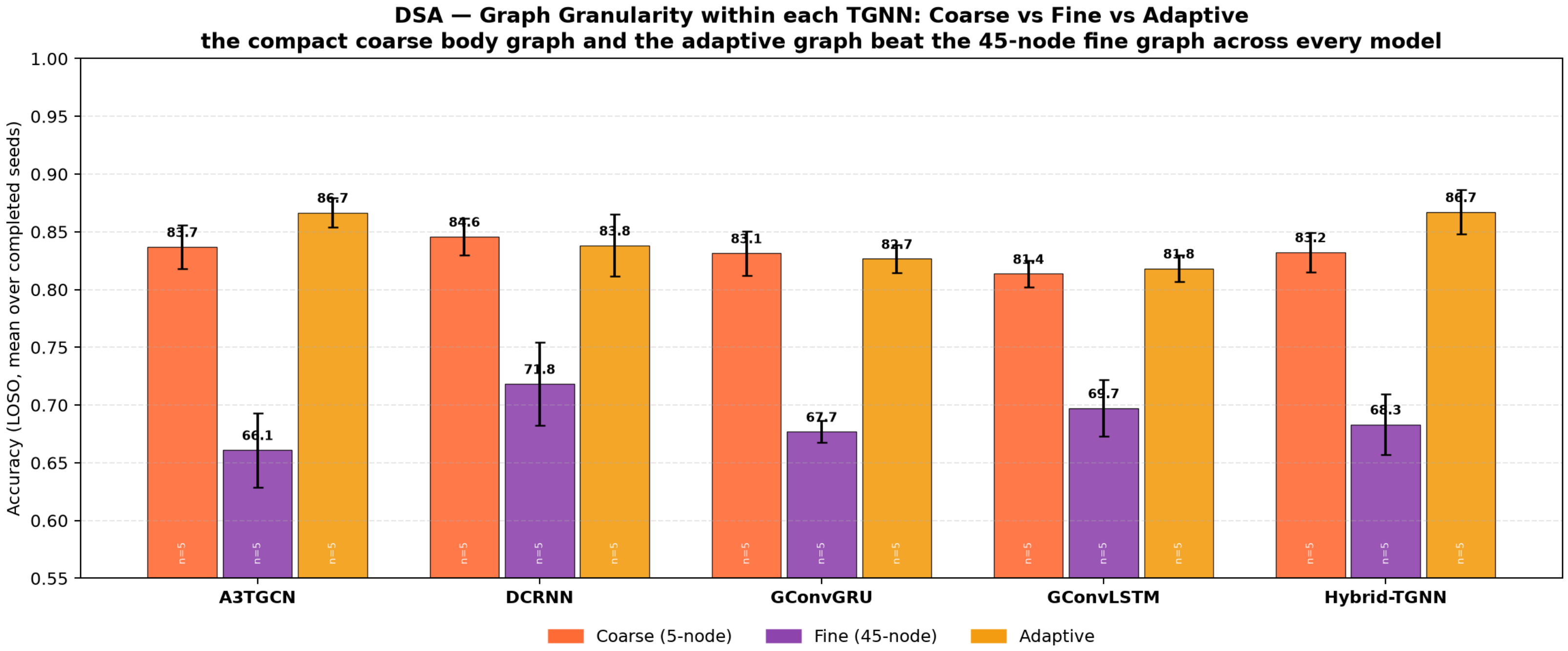
Vertical spread within a model is the graph effect.

Experiment B — 45 channels, 19 classes

Model	Graph	Acc.	Params
A3TGCN	adaptive (5n)	86.7	424 k
Hybrid-TGNN	adaptive (5n)	86.7	620 k
DCRNN	coarse (5n)	84.6	1.21 M
CNN–BiGRU	none	82.9	2.23 M
LSTM	none	79.1	2.33 M
all five TGNNs	fine (45n)	68.7	–

The smallest model is also the most accurate.

Experiment C — is it node granularity?



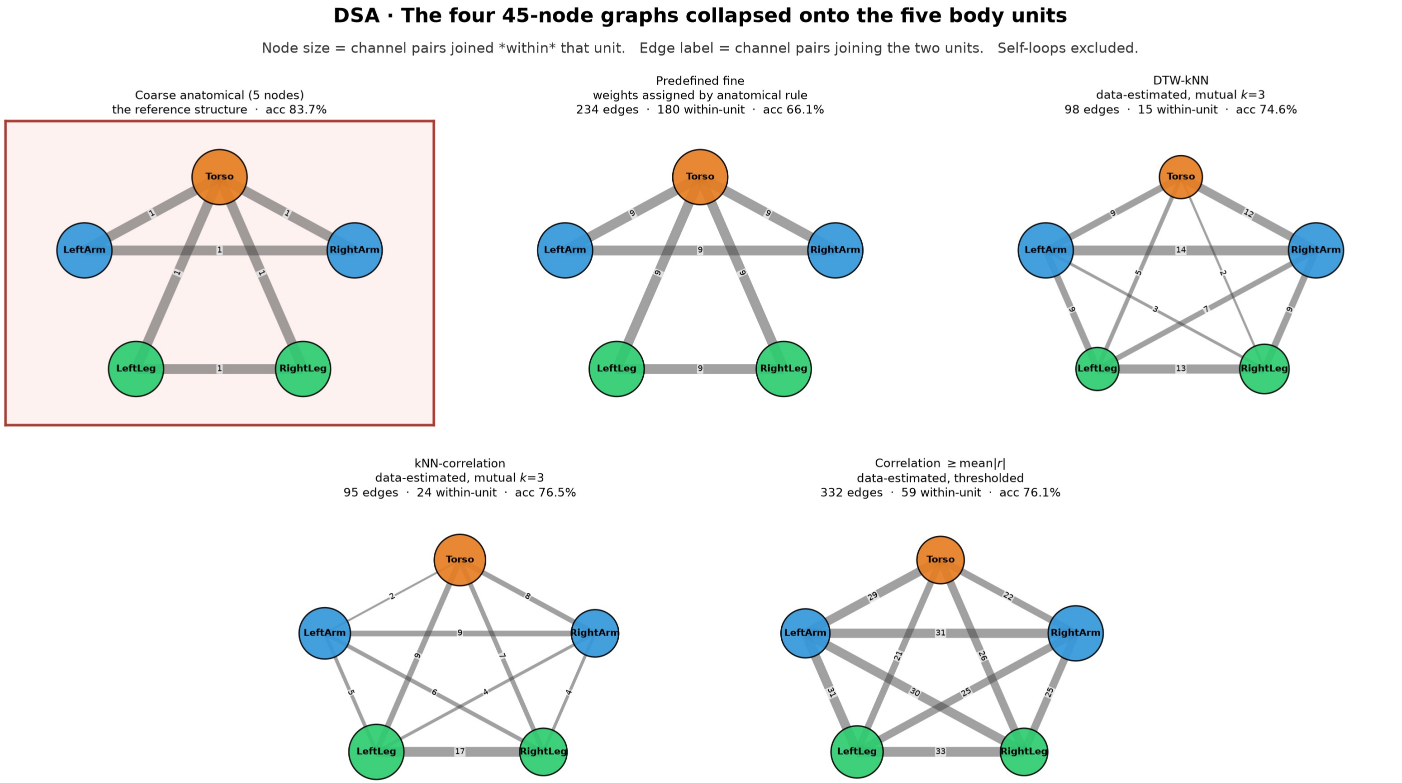
coarse 83.2 fine 68.7 = 14.5 points

The two properties separate cleanly

Graph	Nodes	Weights	Acc.
Adaptive	5	learned	84.3
Predefined coarse	5	uniform prior	83.2
(no graph)	–	–	80.3
Correlation	45	data-estimated	79.7
Predefined fine	45	uniform prior	68.7

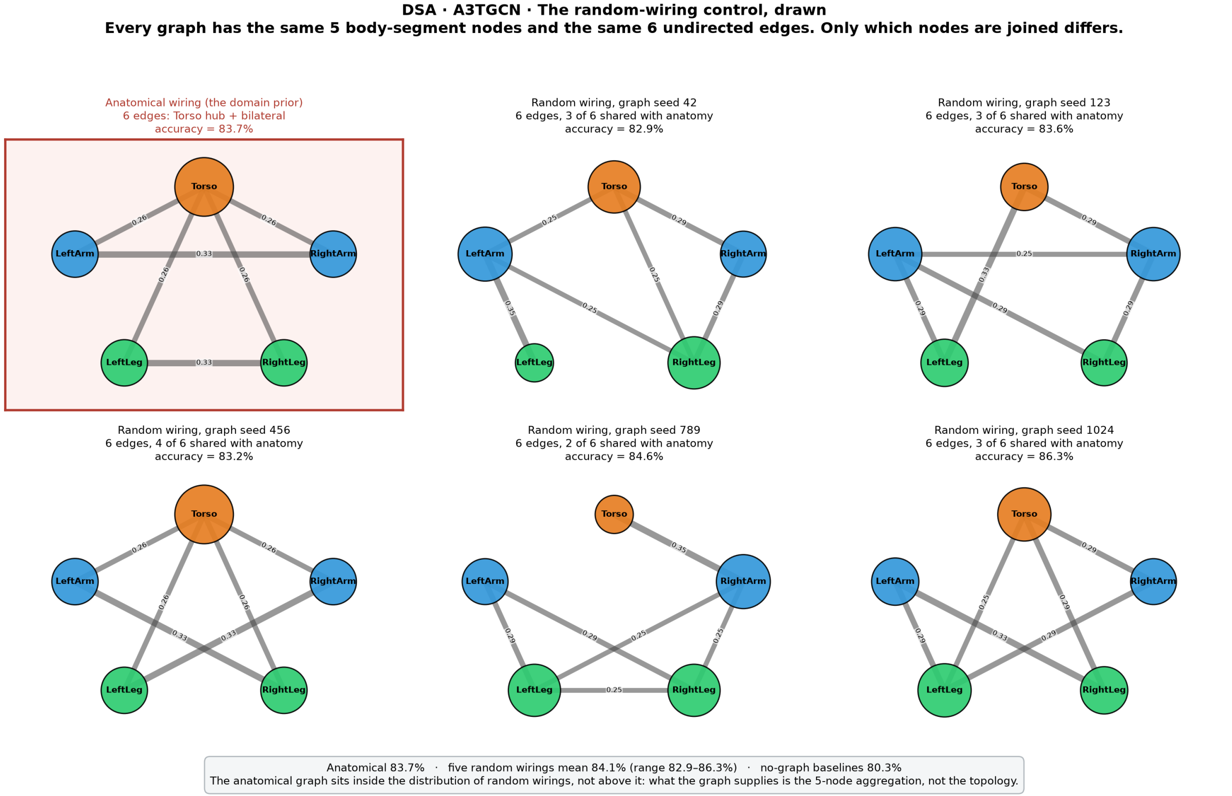
Provenance +11.0

Experiment D — is it edge provenance?



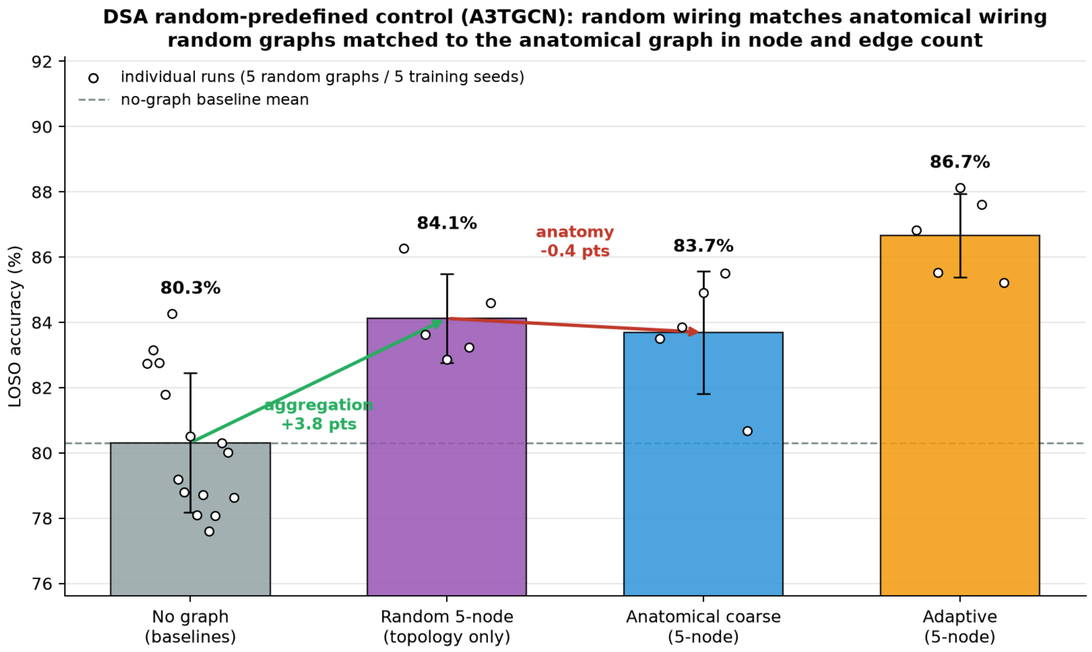
The four 45-node graphs collapsed onto the five body units.

The random topologies, drawn



Same 5 nodes, same 6 edges. Coarse is boxed — and inside the distribution.

Experiment E — does the anatomy contribute?



random 84.1 anatomical 83.7 = −0.4

Random 5-node topologies, matched in node and edge count.

Experiment F — adaptive vs a predefined at 5 nodes

- **Helps for two of five — and those two share a property**

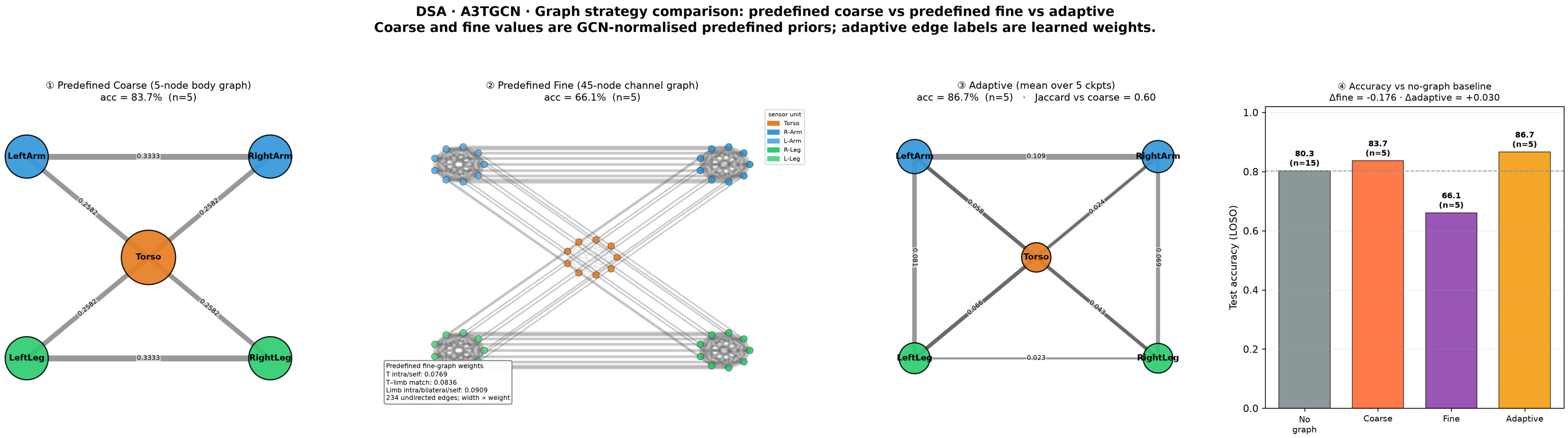
– A3TGCN 83.7 → 86.7 · Hybrid 83.2 → 86.7

– GConvGRU, GConvLSTM, DCRNN: within noise, or behind

- **Both gainers carry an extra selection mechanism**

– attention gate · learned blend

Experiment F — A3TGCN



coarse 83.7 | *fine 66.1* | ***adaptive 86.7***

Experiment G

Nodes	Seed graph	Predefined	Adaptive	Δ
5	anatomical coarse	83.69	86.66	+2.97
45	correlation	76.14	73.03	−3.12
45	DTW–kNN	74.61	71.82	−2.79
45	kNN–correlation	76.47	72.77	−3.70

- **G: granularity decides the SIGN of the learning effect**

Both roles. Two architectures. No effect.

	Adjacency	Anatomical	Random
A3TGCN	Predefined	83.7	84.1
A3TGCN	Adaptive from it	86.7	85.9
GConvLSTM	Predefined	81.4	81.4
GConvLSTM	Adaptive from it	81.8	81.7

- **All four differences under 1 point**
- GConvLSTM chosen for being maximally unlike A3TGCN
- **aggregation +3.8 · making it anatomical −0.4**

Discussion

RQ1
Does the graph help?

- Epilepsy, average from all graph — 94.9 against best baseline 97.2
- Same models, best from graph — all five pass it, 97.8 to 98.8

– the lightest at 105 k against 554 k

- DSA — 86.7 against 82.9, at 424 k parameters against 2.23 M
- Where it costs most, the seizure class, recall moves 0.612 → 0.965
- Graph axis 12.1 to 20.6 · Architecture axis 3.2 to 5.7

RQ2 Which properties?

- Two measured quantities, not a ranking.
- Granularity +14.5 · Provenance +11.0

– every paired observation agrees, without exception

- Correlation-mean is the strongest 45-node graph in the study

– ahead of DTW, kNN, and every learned graph at that granularity

- Five body units genuinely co-move, so estimated weights beat a uniform prior
- Still short of the 5-node graph — aggregation outranks weighting

RQ3
Does the knowledge matter?

- No. And this is a control, not a comparison.
- As a graph — random 84.1 vs anatomy 83.7
- As an initialisation — random 85.9 vs anatomy 86.7
- Aggregation +3.8 over no graph · making it anatomical −0.4
- At 45 channels, partitioning into 5 units beats not partitioning

Conclusions

Conclusions

- Graph construction outranks every other design decision measured here
  - more than the operator, more than whether it is learned
- A sensor graph supplies an aggregation, not a topology
- For practitioners
  - fix the level of aggregation first, then estimate weights from data
- Where the graph is large, initialisation matters. Where it is small, nothing does

Thank you
Any Questions?

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Appendix

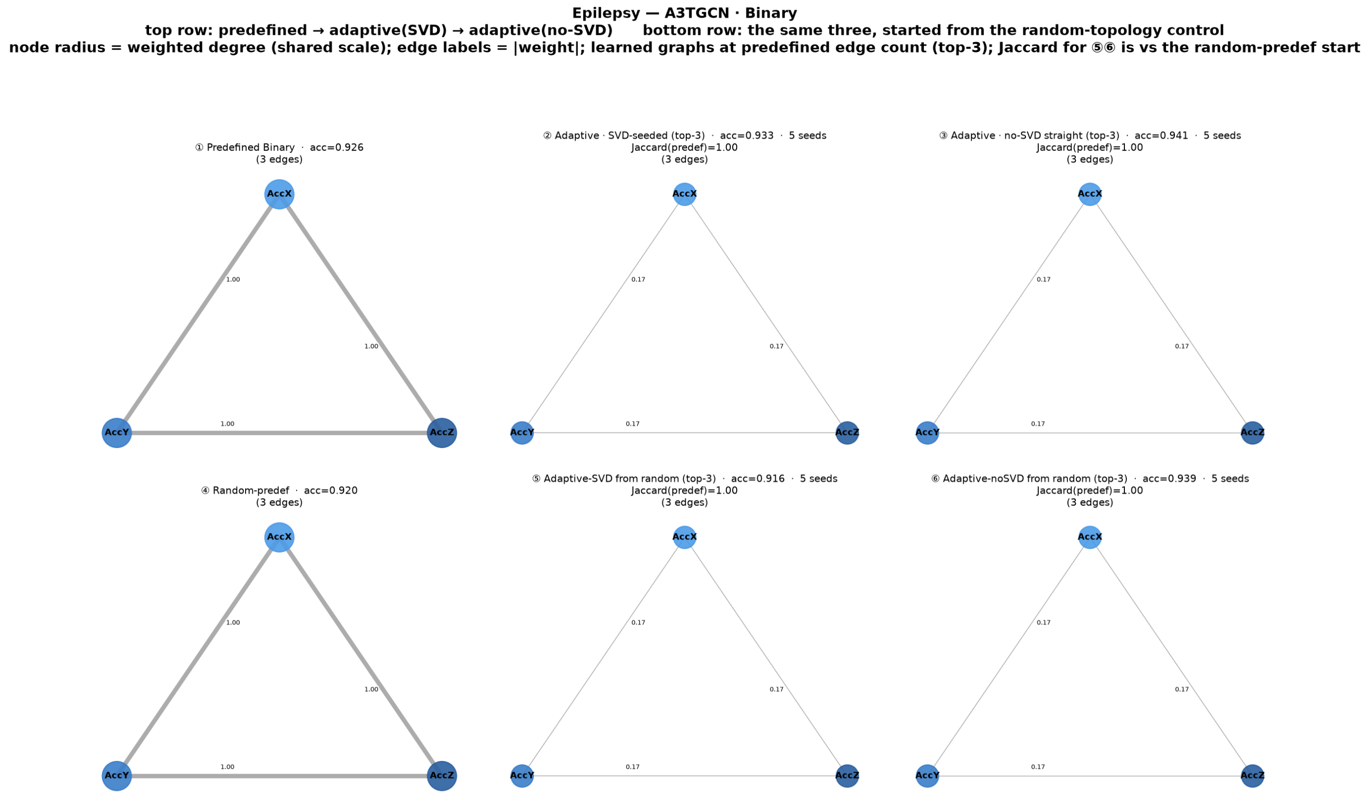
Scale, and what makes it controlled

	Phase 1	Phase 2
Graph modes	41	21
Configurations	405	49
Trained models	1,429	1,320

- 5 seeds · identical preprocessing, optimiser, schedule, metrics
- Every comparison paired across the 8 held-out subjects

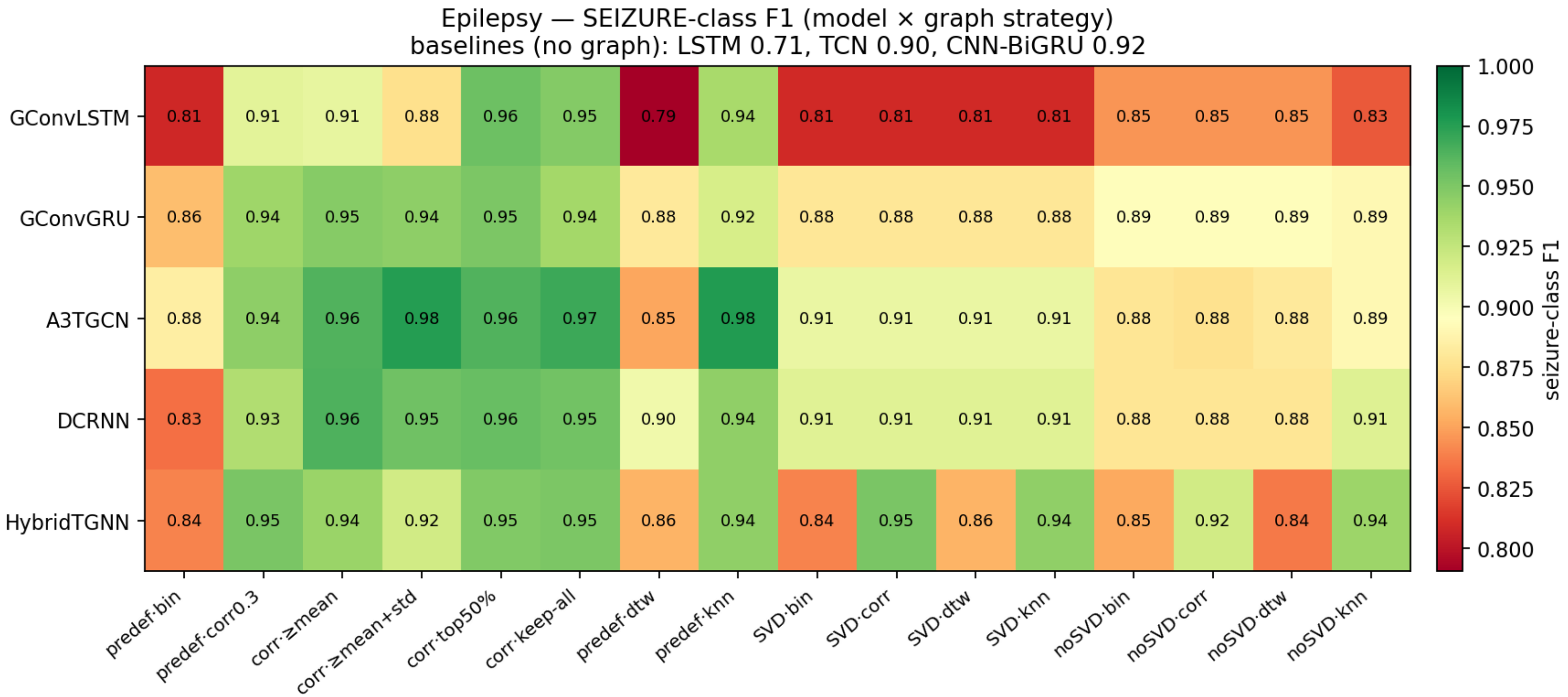
Every comparison paired across the 8 held-out subjects

Epilepsy — predefined axis graph



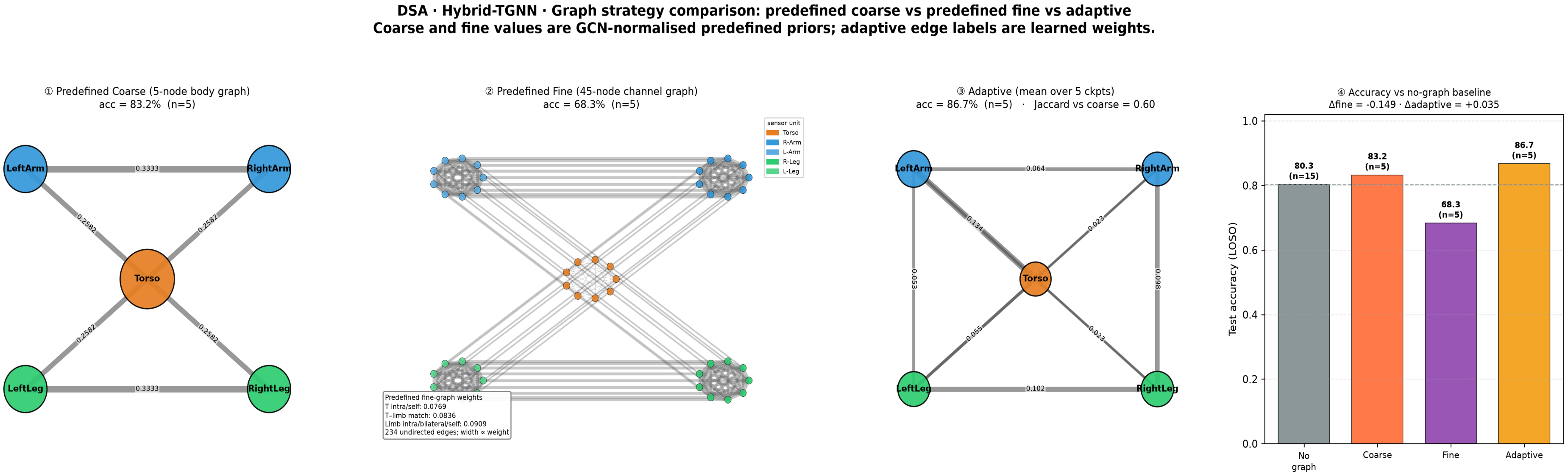
A3TGCN. Three accelerometer axes. Learned graph ≈ its seed.

Seizure-class F1 for every model × strategy



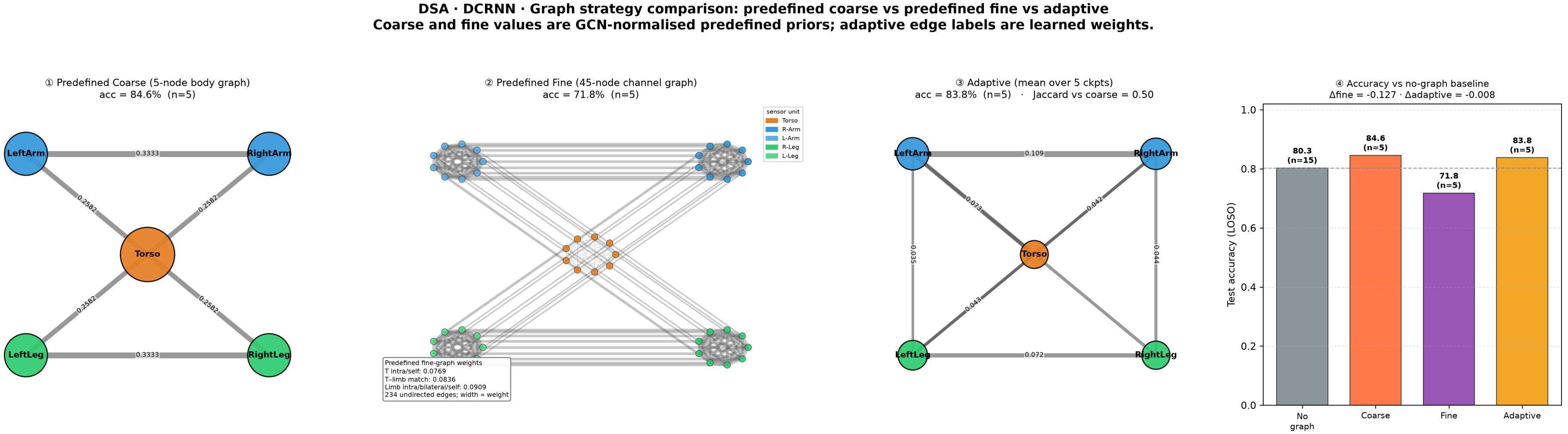
No averaging. Best cell: A3TGCN with the predefined kNN graph, F1 = 0.98.

Experiment F — Hybrid-TGNN



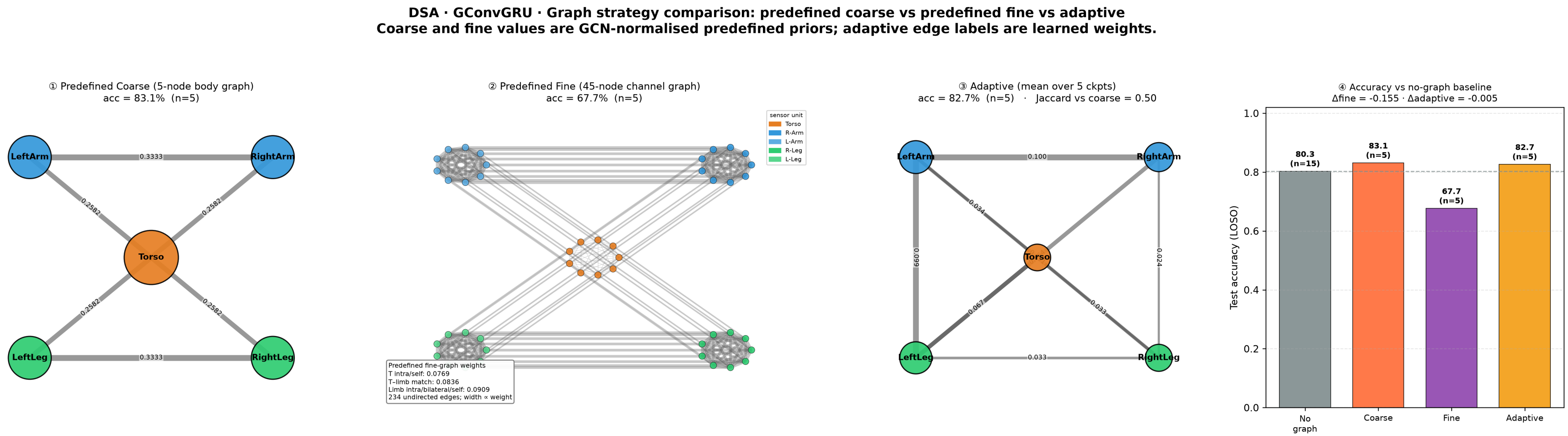
coarse 83.2 | *fine 68.3* | ***adaptive 86.7***

Experiment F — DCRNN



coarse 84.6 | *fine 71.8* | *adaptive 83.8*

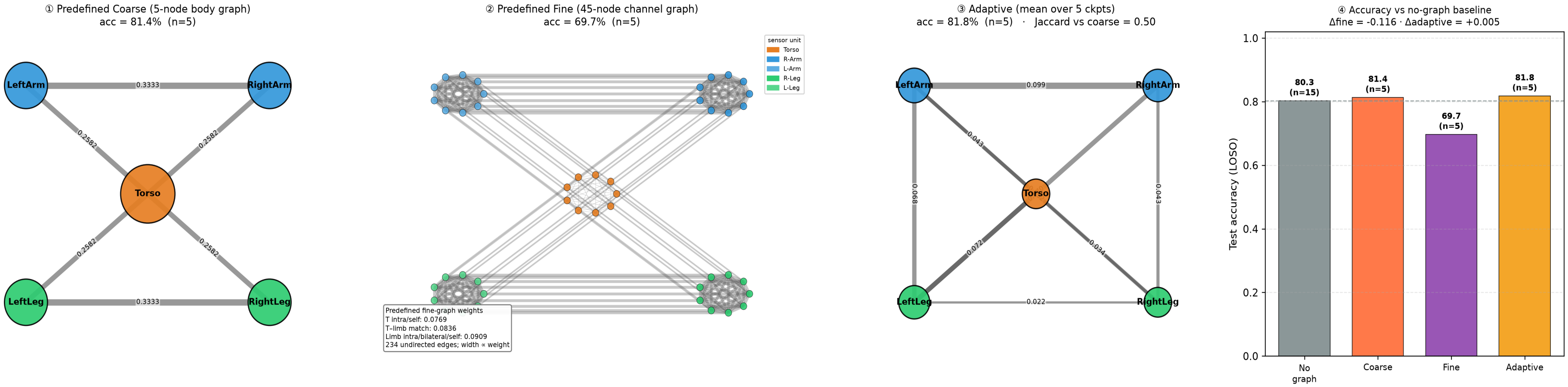
Experiment F — GConvGRU



coarse 83.1 | *fine 67.7* | *adaptive 82.7*

Experiment F — GConvLSTM

DSA · GConvLSTM · Graph strategy comparison: predefined coarse vs predefined fine vs adaptive
Coarse and fine values are GCN-normalised predefined priors; adaptive edge labels are learned weights.



coarse 81.4 | *fine 69.7* | ***adaptive 81.8***

What I defend, and what I do not

- **Established**

– granularity and provenance — every paired observation agrees

– no evidence of an anatomical advantage, in a sensitive design

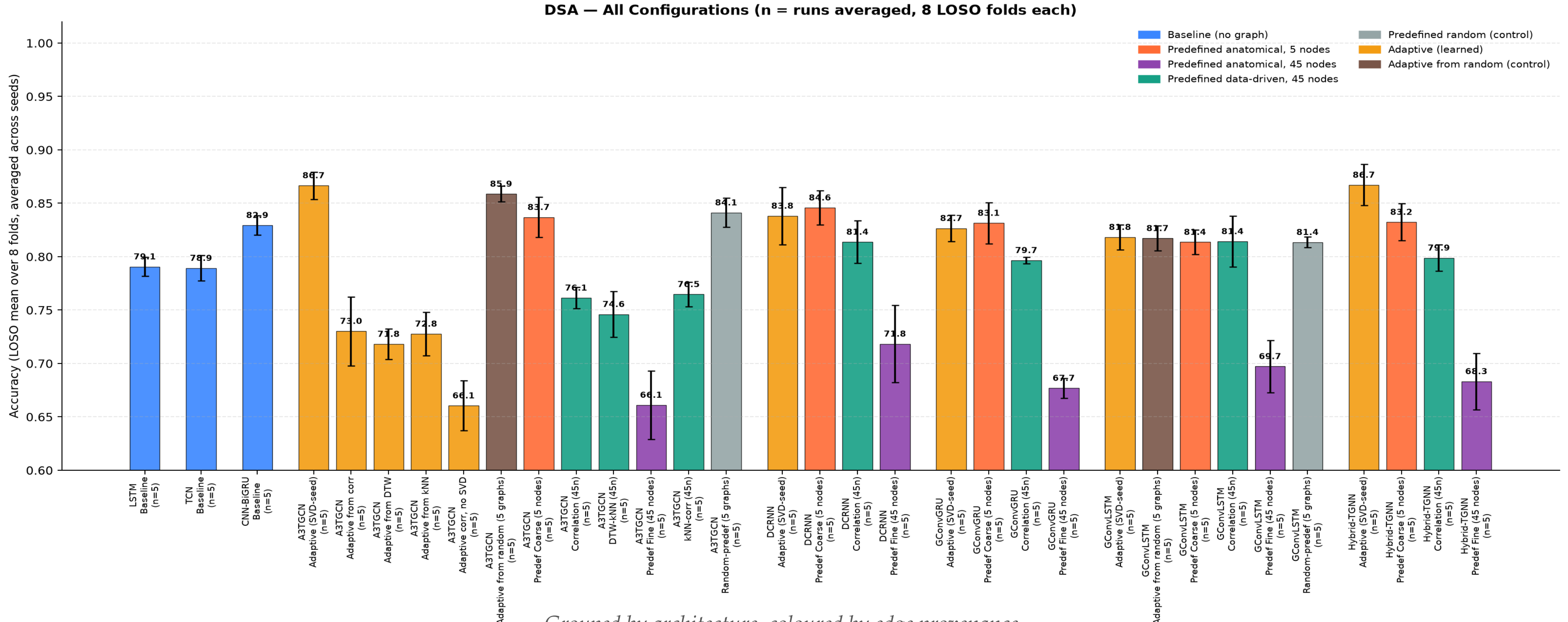
- **Suggestive, not established**

– SVD seeding (+6.98) — large and consistent, but not established at 8 folds

- **Bounded by design**

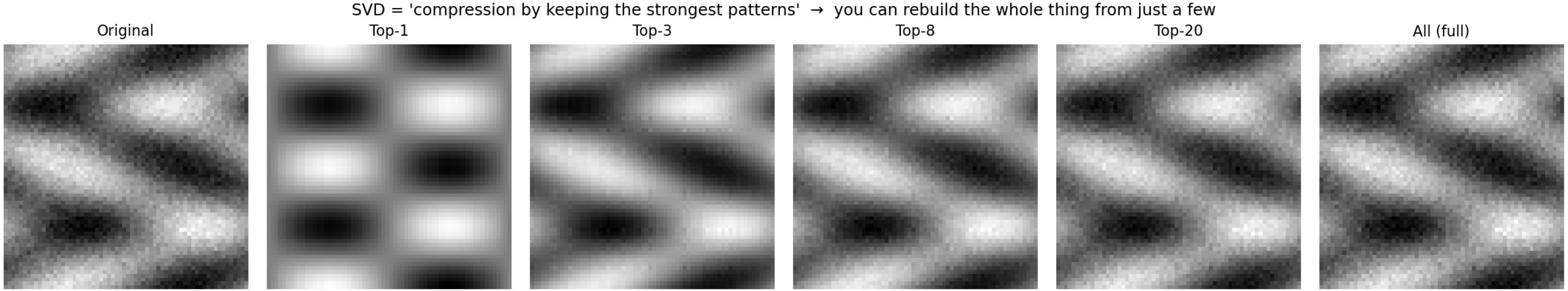
– the control holds at 5 nodes and 6 edges — do not extrapolate

Every configuration in the sweep



Grouped by architecture, coloured by edge provenance.

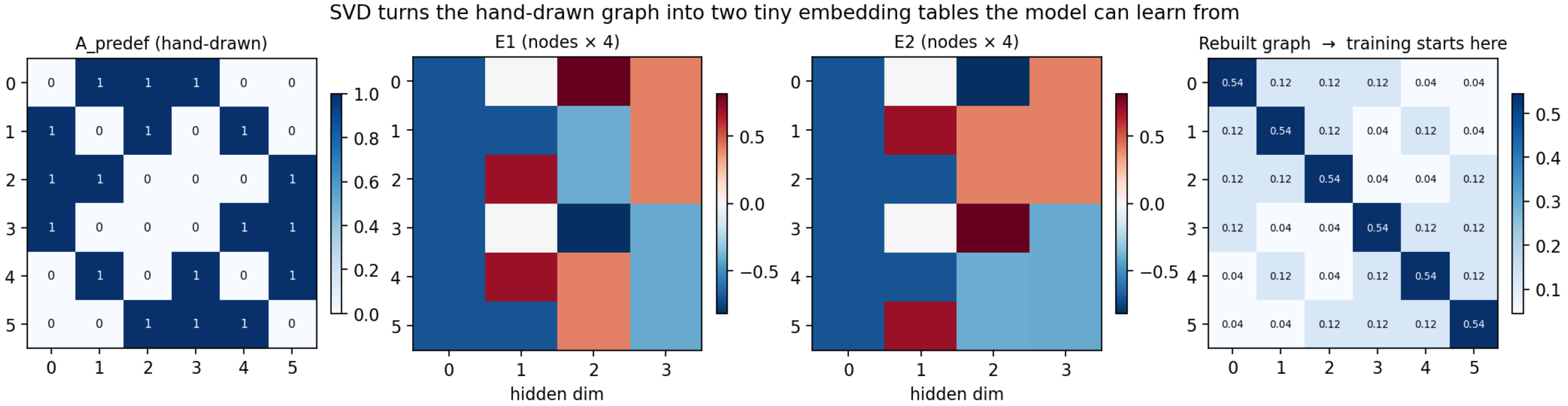
What a truncated SVD does



	Phase 1 · 3 and 6 nodes	Phase 2 · 45 nodes
Paired comparisons	129	5 seeds, A3TGCN
Mean gain from SVD	+0.19	+6.98
Median gain	0.00	+6.98
Better / worse / identical	52 / 29 / 48	5 / 0 / 0
Spread across seeds	–	+4.65 to +11.12

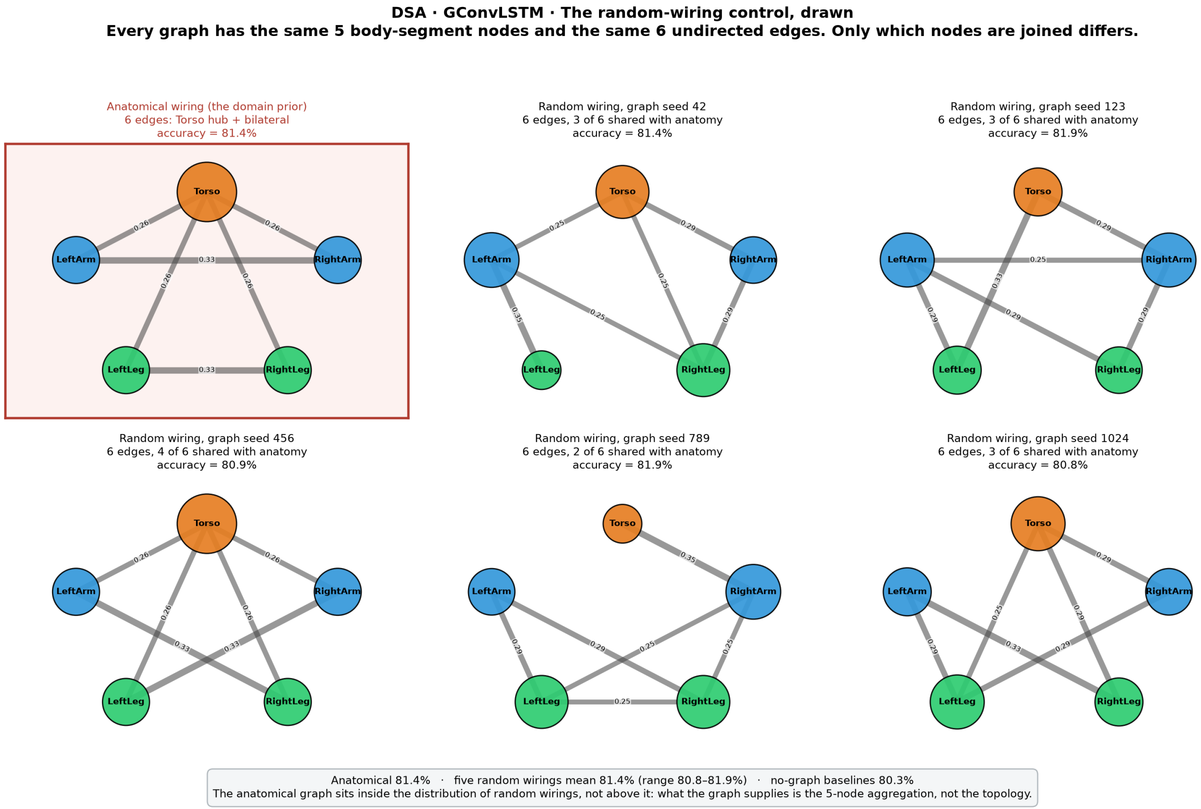
Above, the reconstruction. Below, what the seeding is worth – nothing at 3 and 6 nodes, seven points at 45.

SVD seeding, applied to a graph



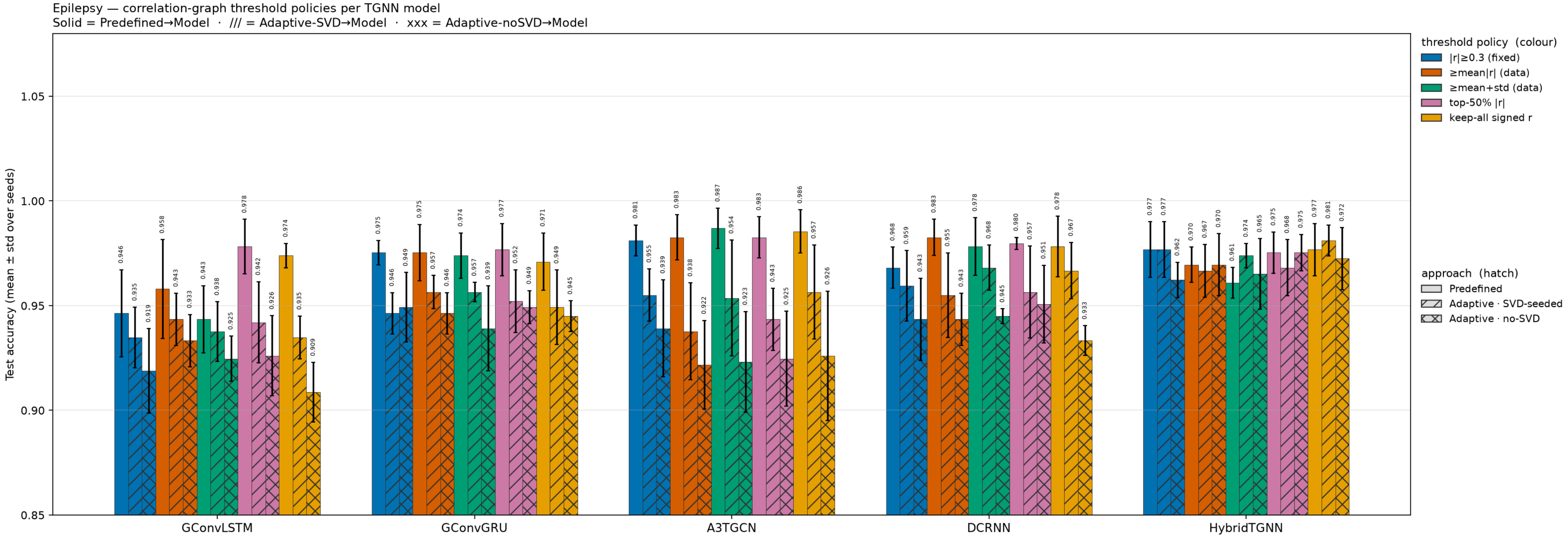
The graph is factorised into the two tables the learner holds.

Random control on GConvLSTM



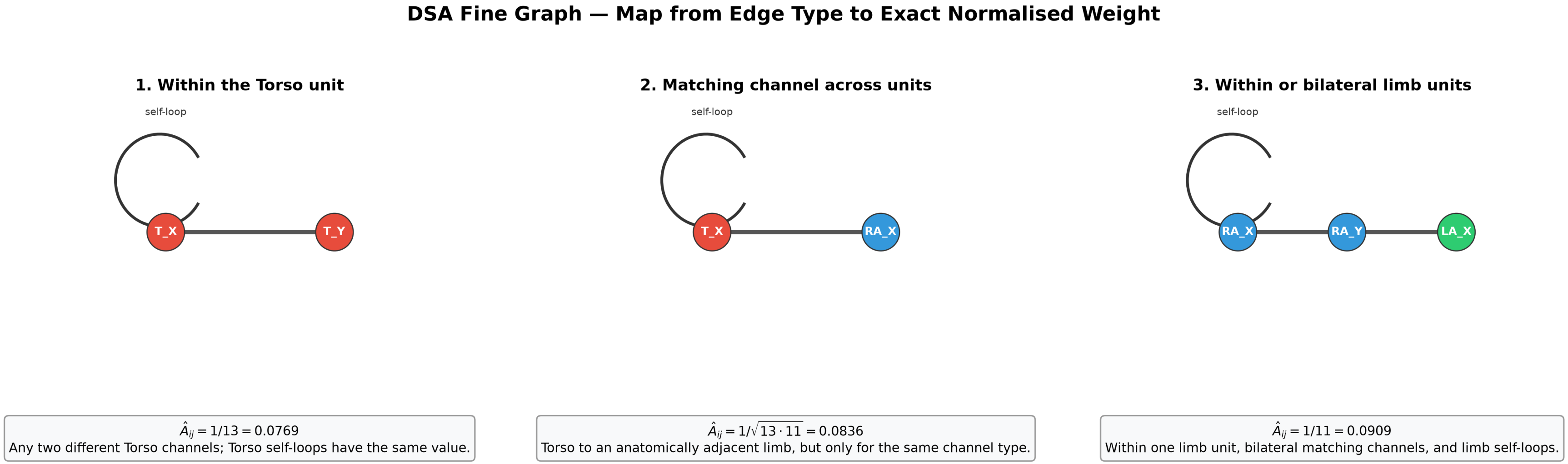
The same five topologies. Anatomy again inside the distribution.

Correlation threshold policies



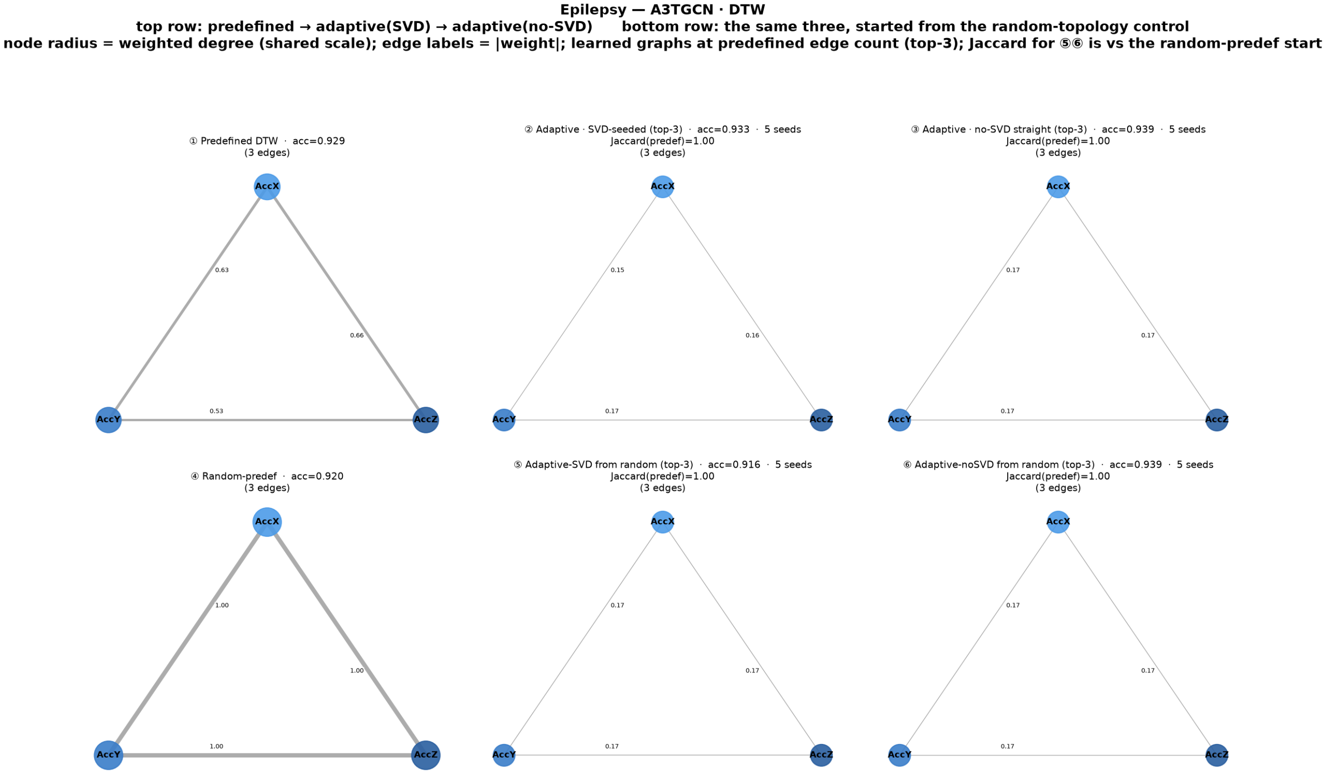
Solid = used directly. Hatched = through the adaptive learner.

Exact weights of the fine graph



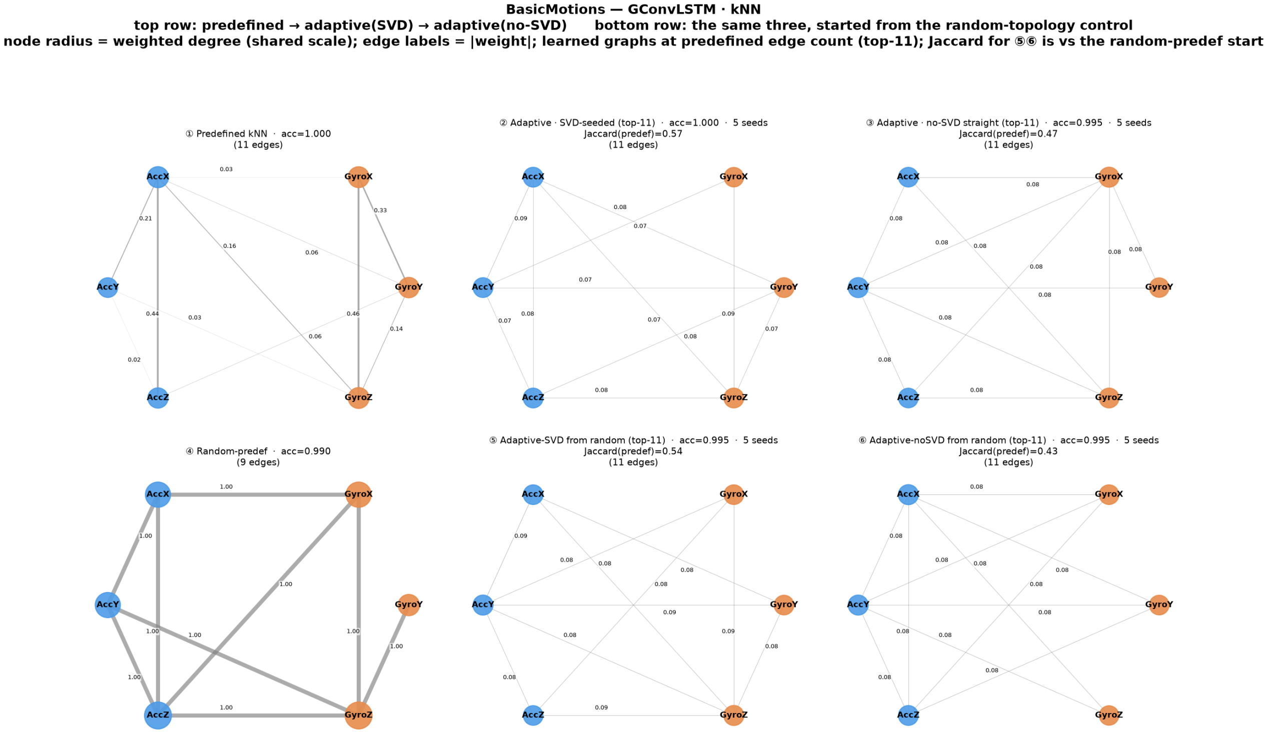
234 edges, THREE distinct values, 18 % end to end.

Epilepsy — DTW graph



The fourth graph source for this architecture.

BasicMotions — kNN graph



The data-driven seed reshapes the six-node graph.

The graph is the larger axis

Vary the GRAPH	Range	Vary the ARCHITECTURE	Range
A3TGCN	20.6	Coarse (5n)	3.2
Hybrid-TGNN	18.4	Adaptive (5n)	4.9
GConvGRU	15.4	Correlation (45n)	5.3
DCRNN	12.8	Fine (45n)	5.7
GConvLSTM	12.1		

The two ranges do not overlap.